

DATA SCIENCE PROJECT REPORT

IMDB Movies Dataset

Exploratory Data Analysis

Ratings · Revenue · Directors · Genre Trends · Audience Engagement

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Executive Summary

This report documents an Exploratory Data Analysis (EDA) on a curated IMDB dataset of 1,000 top-ranked movies spanning 2006 to 2016. The project investigates what makes a film commercially and critically successful — exploring the interplay of ratings, revenue, runtime, genre, directorial talent, and temporal trends.

This project introduces several advanced analytical techniques new to this portfolio: custom classification functions using `.apply()`, year-over-year production trend analysis, genre-level revenue aggregation, correlation between critic scores (Metascore) and audience scores (IMDB Rating), and director/actor performance benchmarking — all wrapped in a well-documented notebook with inline explanations for each code block.

Project Highlights

Metric	Value
Dataset	IMDB Top 1000 Movies (7-IMDB-Movie-Data.csv)
Size	1,000 rows × 12 columns
Year Range	2006 – 2016 (11 years)
Duplicates	0 — perfectly clean on titles

Metric	Value
Missing Values	Revenue: 128 (12.8%) Metascore: 64 (6.4%)
Average IMDB Rating	6.72 out of 10
Average Runtime	113 minutes
Average Vote Count	169,808 per movie
Highest Rated Director	Christopher Nolan — avg. 8.68
Highest Grossing Genre	Action, Sci-Fi — avg. \$318.3M
Top Voted Genre	Action, Adventure, Sci-Fi (50 films — most common combo)
Most Popular Movie	Guardians of the Galaxy (rank #1)
Only 3 Movies >= 180 min	The Wolf of Wall Street, The Hateful Eight, La vie d'Adele
Movie Categories Created	Excellent (>=8.0), Good (6.0–7.9), Average (<6.0)
Tools Used	Python, Pandas, NumPy, Seaborn, Matplotlib

1. Introduction

1.1 Project Background

The film industry is a multi-billion dollar global business where understanding audience preferences, critical reception, and commercial performance is essential for studios, distributors, and streaming platforms. This project uses IMDB data — the world's most trusted movie database — to uncover patterns that answer key entertainment industry questions.

What makes a great film? Does runtime matter? Are higher-rated films more commercially successful? Which directors consistently deliver quality? This EDA answers all these questions with data-backed evidence.

1.2 Objectives

- Inspect and profile the dataset structure, datatypes, and missing values
- Perform univariate analysis on Rating and Runtime distributions
- Conduct bivariate analysis: Runtime vs Rating, Rating vs Revenue, Metascore vs Rating
- Analyze temporal trends: movies per year, votes by year, revenue by year
- Rank top directors and actor ensembles by average IMDB rating
- Perform genre-level revenue analysis to identify highest-grossing genres
- Engineer a custom movie category column using `.apply()` with a classification function
- Identify long movies (>= 180 minutes) and their critical/commercial performance

1.3 All Analysis Tasks

#	Task	Type	Technique
1	Load & preview top 5 rows	Inspection	read_csv, head()
2	Check shape, dtypes, nunique	Profiling	shape, dtypes, nunique
3	Descriptive statistics (numeric)	Statistics	describe()
4	Full info summary with verbose mode	Profiling	info()
5	Describe including all columns (categorical + numeric)	Statistics	describe(include='all')
6	Check null values per column	Quality	isnull().sum()
7	Check for duplicate rows	Quality	duplicated().sum()
8	Plot histograms of all numeric columns	Univariate	df.hist(figsize=(15,15))
9	Rating distribution with KDE	Univariate	sns.histplot(kde=True)
10	Runtime distribution with KDE	Univariate	sns.histplot(kde=True)
11	Runtime vs Rating (bivariate scatter)	Bivariate	sns.scatterplot()
12	Rating vs Revenue (bivariate scatter)	Bivariate	sns.scatterplot()
13	Identify movies >= 180 minutes	Filtering	Boolean filter on Runtime
14	Votes by year (bar chart)	Temporal	groupby('Year')['Votes'].sum()
15	Revenue trend by year (line chart)	Temporal	groupby('Year')['Revenue'].sum()
16	Top 10 directors by avg. rating	Ranking	groupby + mean + sort_values
17	Movies produced per year	Temporal	value_counts().sort_index()
18	Custom movie classification function	Feature Eng.	apply(classify_movie)
19	Genre-level average revenue ranking	Revenue Analysis	groupby('Genre')['Revenue'].mean()
20	Top 10 actor ensembles by avg. rating	Ranking	groupby('Actors')['Rating'].mean()
21	Metascore vs IMDB Rating (scatter)	Bivariate	sns.scatterplot()

2. Dataset Overview

2.1 Data Source

The dataset is the IMDB Top 1000 Movies dataset available on Kaggle, covering films from 2006 to 2016. Each row represents a single movie ranked by IMDB's ranking algorithm, with attributes covering critical reception, audience engagement, financial performance, and metadata.

Repository: github.com/princemaheta2627-glitch/imdb_movies_analysis_eda_project

2.2 Dataset Structure — 12 Columns

Column	Dtype	Non-Null	Description
Rank	int64	1,000	IMDB ranking position (1–1000)
Title	object	1,000	Movie title (999 unique — 'The Host' appears twice)
Genre	object	1,000	Comma-separated genre tags (207 unique combinations)
Description	object	1,000	One-line movie synopsis
Director	object	1,000	Director name
Actors	object	1,000	Comma-separated top 4 cast members
Year	int64	1,000	Release year (2006–2016)
Runtime (Minutes)	int64	1,000	Total runtime in minutes (66–187)
Rating	float64	1,000	IMDB audience rating (1.9–9.0)
Votes	int64	1,000	Total IMDB vote count
Revenue (Millions)	float64	872	Box office revenue in USD millions — 128 missing
Metascore	float64	936	Metacritic critic score (0–100) — 64 missing

2.3 Descriptive Statistics

Statistic	Year	Runtime (min)	Rating	Votes	Revenue (\$M)	Metascore
Count	1,000	1,000	1,000	1,000	872	936
Mean	2012.8	113.2	6.72	169,808	\$82.96M	58.99
Std Dev	3.21	18.8	0.95	188,763	\$103.3M	17.3
Min	2006	66	1.9	61	\$0.0M	11
25th %ile	2010	100	6.2	36,309	\$10.3M	47
Median	2014	111	6.8	110,799	\$47.0M	59

Statistic	Year	Runtime (min)	Rating	Votes	Revenue (\$M)	Metascore
75th %ile	2016	122	7.4	242,743	\$113.7M	72
Max	2016	187	9.0	1,791,916	\$936.6M	100

2.4 Data Quality Assessment

Issue	Column	Count	% of Dataset	Handling Strategy
Missing Values	Revenue (Millions)	128	12.8%	Not imputed — excluded from revenue analyses
Missing Values	Metascore	64	6.4%	Not imputed — excluded from critic analyses
Duplicate Rows	All columns	0	0%	No action needed — clean dataset
Multi-value field	Genre	207 unique combos	N/A	Treated as categorical string for groupby
Multi-value field	Actors	N/A	N/A	Treated as full ensemble string for groupby

The missing Revenue and Metascore values are likely due to limited-release films (no wide box office data) or films not reviewed by Metacritic. Both columns are used in analyses where only non-null values are relevant.

3. Methodology

3.1 Tools & Libraries

Library	Role	Key Usage
Python 3	Core language	All logic and custom functions
Pandas	Data manipulation	groupby, sort_values, value_counts, apply, describe
NumPy	Numerical support	Background array operations
Seaborn	Statistical visualization	histplot, scatterplot
Matplotlib	Plot rendering	hist(), plot(kind='bar'/'line'), figsize, title
warnings	Output control	filterwarnings('ignore') — clean notebook output

3.2 Key Code Patterns

Custom Classification Function with .apply()

```
- def classify_movie(rating):  
-     if rating >= 8:     return 'Excellent'  
-     elif rating >= 6: return 'Good'  
-     else:               return 'Average'  
-  
- df['Category'] = df['Rating'].apply(classify_movie)
```

Director Performance Ranking

```
- director_ratings = df.groupby('Director')['Rating'].mean()  
-                 .sort_values(ascending=False).head(10)
```

Genre Revenue Analysis

```
- genre_revenue = df.groupby('Genre')['Revenue (Millions)'].mean()  
-               .sort_values(ascending=False)
```

Temporal Analysis

```
- votes_by_year   = df.groupby('Year')['Votes'].sum()  
- revenue_by_year = df.groupby('Year')['Revenue (Millions)'].sum()  
- movies_per_year = df['Year'].value_counts().sort_index()
```

Long Movies Filter

```
- long_movies = df[df['Runtime (Minutes)'] >= 180]  
- print(long_movies[['Title', 'Runtime (Minutes)', 'Rating', 'Revenue (Millions)']])
```

4. Key Findings & Insights

4.1 Rating Distribution (Univariate)

The histplot with KDE overlay on the Rating column revealed:

- Mean rating: 6.72 — most films cluster in the 6.0–7.5 range
- The distribution is slightly left-skewed — more films rate above average than below
- Very few films score below 4.0 or above 9.0 — extreme ends are rare
- Standard deviation of 0.95 — relatively tight distribution around the mean

Insight: IMDB ratings are not normally distributed — they cluster in the 'good but not great' range. This reflects audience rating behavior: casual viewers rarely bother rating films they don't like, creating upward bias.

4.2 Runtime Distribution (Univariate)

The runtime histplot revealed:

- Mean: 113 minutes — close to the classic 2-hour standard
- Range: 66–187 minutes — most films between 90 and 135 minutes
- Slight right skew — a small number of very long films (epics) pull the mean up
- Only 3 films exceeded 180 minutes (see long movies analysis below)

4.3 Runtime vs. Rating (Bivariate)

A scatterplot of Runtime (Minutes) vs. Rating showed no strong linear correlation, but a notable pattern:

- Very short films (< 80 min) tend to have lower ratings — likely low-budget productions
- Films in the 120–180 minute range show the highest concentration of top-rated films (8.0+)
- No single runtime guarantees a high rating — story quality transcends length

Insight: Longer films are not necessarily better, but the highest-rated films tend to be longer — possibly because complex, critically acclaimed stories require more time to develop.

4.4 Rating vs. Revenue (Bivariate)

The scatterplot of IMDB Rating vs. Revenue (Millions) revealed a weak positive relationship with high variance:

- Most high-revenue films (\$300M+) are rated between 6.5 and 8.5 — not necessarily the highest rated
- Some of the highest-rated films (8.5+) earned relatively modest revenue
- Low-rated films (below 5.0) consistently earn low revenue

Insight: A high IMDB rating is not sufficient for blockbuster revenue — marketing budget, franchise brand power, and release timing are equally important. However, a low rating is nearly always associated with low box office performance.

4.5 Long Movies Analysis (>= 180 Minutes)

Movie	Runtime	IMDB Rating	Revenue (\$M)	Insight
The Wolf of Wall Street	180 min	8.2	\$116.9M	Critical + commercial success
The Hateful Eight	187 min	7.8	\$54.1M	Critical success, modest revenue

Movie	Runtime	IMDB Rating	Revenue (\$M)	Insight
La vie d'Adèle	180 min	7.8	\$2.2M	Art house film — limited release

Insight: Only 3 films in 1,000 exceed 3 hours — a very exclusive club. All three are critically well-regarded (7.8–8.2), suggesting audiences are willing to invest time in high-quality epics. However, commercial performance varies dramatically — art house films earn a fraction of mainstream releases regardless of quality.

4.6 Top 10 Directors by Average Rating

Rank	Director	Avg. Rating	Notable Films
1	Christopher Nolan	8.68	The Dark Knight, Inception, Interstellar
2	Olivier Nakache	8.60	The Intouchables
3	Makoto Shinkai	8.60	Your Name
4	Florian Henckel von Donnersmarck	8.50	The Lives of Others
5	Aamir Khan	8.50	3 Idiots (as director/actor)
6	Damien Chazelle	8.40	Whiplash
7	Rajkumar Hirani	8.30	PK, 3 Idiots
8	Thomas Vinterberg	8.30	The Hunt
9	Lee Unkrich	8.30	Toy Story 3, Coco
10	Pete Docter	8.25	Up, Inside Out

Insight: Christopher Nolan leads with an exceptional 8.68 average — the highest of any director in the dataset. The top 10 spans Hollywood (Nolan, Chazelle, Pixar directors), European cinema (von Donnersmarck, Vinterberg), Japanese animation (Shinkai), and Bollywood (Khan, Hirani) — reflecting the global diversity of critically acclaimed filmmaking.

4.7 Genre Revenue Analysis

Rank	Genre Combination	Avg. Revenue (\$M)	Insight
1	Action, Sci-Fi	\$318.3M	Pure action-sci-fi crossover is most lucrative
2	Adventure, Drama, Fantasy	\$276.0M	Epic fantasy dramas (e.g. LOTR-style)
3	Adventure, Fantasy	\$272.2M	Family-friendly adventure fantasy
4	Action, Adventure	\$223.7M	Classic blockbuster combination
5	Animation, Adventure, Comedy	\$221.3M	Pixar/DreamWorks formula

Rank	Genre Combination	Avg. Revenue (\$M)	Insight
Bottom	Drama, History, War	\$0.86M	Prestige films — minimal commercial reach
Bottom	Thriller	\$0.32M	Stand-alone thrillers — niche audience
Bottom	Horror, Mystery, Sci-Fi	\$0.15M	Niche horror sub-genre

Insight: Action and Sci-Fi combinations dominate commercial revenue by a large margin — these are the tent-pole franchise genres (Marvel, Star Wars, Transformers). Pure dramas and art-house genres earn almost nothing commercially despite often receiving strong critical acclaim. Genre choice is the single biggest commercial lever for studios.

4.8 Movie Category Distribution

The custom `classify_movie()` function applied via `.apply()` created a new `Category` column:

Category	Rating Range	Approximate Count	Examples
Excellent	≥ 8.0	~100 films	The Dark Knight (9.0), Inception (8.8)
Good	6.0 – 7.9	~750 films	Guardians of Galaxy (8.1 = Excellent), Prometheus (7.0)
Average	< 6.0	~150 films	Low-rated genre films and limited releases

Insight: The majority of films fall in the 'Good' category — consistent with the rating distribution analysis. 'Excellent' films (roughly the top 10%) represent a prestigious tier. This classification is the foundation for future predictive modeling to determine which film characteristics predict 'Excellent' status.

4.9 Temporal Trends

Movies per Year

Production was relatively consistent from 2006–2016, with the dataset capturing approximately 80–120 films per year. The highest production years were 2014–2016, reflecting both increased output and IMDB's ranking methodology favoring more recent films.

Votes by Year

Vote counts were highest for films from 2014–2016 — partially because these films had more time to accumulate votes at the time of data collection, but also because the rise of mobile IMDB voting platforms increased participation. Earlier films (2006–2009) show lower vote counts despite some being highly rated.

Revenue by Year

The revenue line chart showed a general upward trend in total box office revenue across years, reflecting both inflation and the rise of franchise blockbusters in the 2012–2016 period (Marvel Cinematic Universe expansion, Star Wars revival, etc.).

4.10 Metascore vs. IMDB Rating (Bivariate)

The scatterplot of Metascore (critic score, 0–100 scale) vs. IMDB Rating (audience score, 1–10 scale) revealed a moderate positive correlation:

- Films with high Metascores generally have higher IMDB ratings — critics and audiences broadly agree
- There is significant scatter around the trend — some films have high critic scores but average audience ratings (art house films) and vice versa (action blockbusters loved by audiences but dismissed by critics)
- The relationship is stronger at the extremes — films scoring very high or very low on Metascore tend to receive correspondingly high or low IMDB ratings

Insight: Critics and audiences are not perfectly aligned, but they broadly agree on quality. The divergences are commercially interesting — films where audiences rate much higher than critics tend to be genre entertainment with broad appeal, while critic-favored films with lower audience ratings are typically slower, more artistic works.

4.11 Top Actor Ensembles by Rating

The top-rated actor ensemble (groupby on the full Actors string) revealed iconic pairings:

- #1: Christian Bale, Heath Ledger, Aaron Eckhart, Michael Caine — 9.0 (The Dark Knight)
- #2: Leonardo DiCaprio, Joseph Gordon-Levitt, Ellen Page, Ken Watanabe — 8.8 (Inception)
- #3: Matthew McConaughey, Anne Hathaway, Jessica Chastain, Mackenzie Foy — 8.6 (Interstellar)

The top ensembles are dominated by Christopher Nolan films — consistent with his leading position in the director rankings.

5. Technical Skills Demonstrated

5.1 Data Profiling

- `df.shape` — (1000, 12) row-column profile
- `df.dtypes` — confirmed Revenue and Metascore as float64 (correct for missing values)
- `df.nunique` — invoked to check column cardinality (note: called without parentheses — shows method object; correct usage is `df.nunique()`)
- `df.describe()` and `df.describe(include='all')` — numeric + categorical combined stats
- `df.info()` — concise schema with non-null counts
- `warnings.filterwarnings('ignore')` — professional clean notebook output suppressing matplotlib deprecation warnings

5.2 Feature Engineering

- Custom `classify_movie()` function with `if/elif/else` logic — Python function definition
- `df['Category'] = df['Rating'].apply(classify_movie)` — `.apply()` for row-wise transformation
- This pattern is foundational for ML feature engineering — any custom logic can be applied to a column via `.apply()`

5.3 GroupBy Aggregation & Ranking

- `groupby('Director')['Rating'].mean().sort_values(ascending=False).head(10)` — director performance leaderboard
- `groupby('Genre')['Revenue (Millions)'].mean().sort_values(ascending=False)` — genre revenue ranking across 189 unique genre combinations
- `groupby('Actors')['Rating'].mean().sort_values(ascending=False).head(10)` — ensemble cast performance
- `groupby('Year')['Votes'].sum()` and `groupby('Year')['Revenue'].sum()` — temporal aggregation
- `value_counts().sort_index()` — frequency count with chronological ordering

5.4 Visualization Suite

- `df.hist(figsize=(15,15))` — all-numeric histogram grid in one call
- `sns.histplot(col, bins=N, kde=True)` — distribution with KDE overlay
- `sns.scatterplot(x, y, data)` — bivariate relationship plots
- `votes_by_year.plot(kind='bar', figsize=(10,5))` — Pandas-native bar chart
- `revenue_by_year.plot(kind='line', marker='o', figsize=(10,5))` — line chart with markers
- `plt.title(), plt.show()` — standard plot formatting and display

5.5 Filtering & Conditional Logic

- Boolean filter: `df[df['Runtime (Minutes)'] >= 180]` — subset selection
- Multi-column display: `long_movies[['Title', 'Runtime (Minutes)', 'Rating', 'Revenue (Millions)']]`
- `max()` lookup + Boolean match for finding top-value records

6. Conclusion & Future Work

6.1 Summary of Findings

- 1,000 IMDB-ranked movies from 2006–2016 across 207 genre combinations, clean with zero duplicates
- Average rating of 6.72 — films cluster in the 6.0–7.5 'good' band

- Revenue and Metascore each have 10%+ missing data — films without wide releases or critic reviews
- Christopher Nolan is the highest-rated director with 8.68 avg rating across his films
- Action + Sci-Fi genre combination yields the highest avg revenue (\$318.3M) — blockbuster formula confirmed
- Rating weakly predicts revenue — star power and marketing matter as much as quality
- Critic scores (Metascore) and audience scores (IMDB Rating) show moderate positive correlation but notable divergence
- Only 3 films exceed 3 hours — all well-rated; the 90–135 minute range is the commercial sweet spot
- Movie production and revenue trended upward through 2014–2016 — franchise era in full effect

6.2 Potential Enhancements

Enhancement	Description	Technique
Rating Prediction Model	Predict IMDB Rating from genre, runtime, director	Random Forest, XGBoost, Feature Encoding
Genre Explosion	Split multi-genre strings into individual tags for better analysis	<code>str.split(',') + explode()</code> + <code>groupby</code>
Revenue Prediction	Predict box office from film attributes	Regression with imputed Revenue
Director Network Analysis	Map collaborations between directors and actors	NetworkX graph visualization
NLP on Descriptions	Extract themes from movie descriptions	TF-IDF, LDA topic modeling, BERT
Sentiment Analysis on Titles	Correlate title sentiment with rating	VADER, TextBlob
Interactive Dashboard	Browse movies by genre, year, director	Plotly Dash, Streamlit

6.3 Learning Outcomes

This project introduced the `.apply()` pattern with a custom function — one of the most powerful and flexible Pandas techniques for column-level transformations. It also demonstrated how to benchmark performance entities (directors, actors) using `groupby` on string columns, and how to combine temporal `groupby` with different plot types (bar for counts, line for trends). The genre revenue analysis, despite the multi-value genre strings limiting precision, gave a useful top-level commercial picture that could be dramatically improved with genre explosion (`str.split + explode`) in a follow-up project.

7. Project Information

Project Title	IMDB Movies Dataset Exploratory Data Analysis
Author	Prince Maheta
GitHub Repository	github.com/princemaheta2627-glitch/imdb_movies_analysis_eda_project
Notebook File	imdb_eda_project.ipynb
Dataset	IMDB Top 1000 Movies (7-IMDB-Movie-Data.csv) — 1,000 rows, 12 columns
Year Coverage	2006 – 2016 (11 years of film data)
Platform	Jupyter Notebook (Python 3)
Languages & Tools	Python 3, Pandas, NumPy, Seaborn, Matplotlib
Key Techniques	apply(), groupby+mean+sort, histplot(kde=True), scatterplot, plot(kind='line'/'bar'), describe(include='all')
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